Hongrui (Jimmy) Shi

Postdoctoral Research Associate at the University of Lincoln. Builds and evaluates computer-vision systems for UK habitat monitoring, with a focus on when their predictions can be trusted.

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EXPERIENCE

Sep 2024 - **Postdoctoral Research Associate**

University of Lincoln · Lincoln, UK

- Lead AI model development for AI-Hab, a NERC-funded University of Lincoln-UKCEH collaboration; curated an analytical dataset of over 5,000 expert-labelled photographs across 18 fine-grained UKHab-aligned habitat classes from a source pool exceeding 10,000 UKCEH Countryside Survey images.
- Designed reproducible training and evaluation pipelines comparing nine convolutional and vision-transformer models; the best reached 71.5% top-1 and 90.9% top-3 accuracy, comparable to the strongest of three experienced ecologists on a stratified held-out subset.
- Led the formal analysis and first-author paper in Ecological Informatics, examining class imbalance and habitat-level errors; contrastive learning raised class-specific recall by six percentage points for Improved Grassland and five for Neutral Grassland.
- Extended AI-Hab beyond image-only assessment with a multimodal system combining imagery, geospatial data (Google Satellite Embeddings) and LLM-derived ecological descriptions; internal results show a two-point top-1 gain over the image-only baseline.
- Released public code, trained models and a public demonstrator; co-organised the 2025 stakeholder workshop involving Defra and Mozaic Earth.

Sep 2024 - Apr 2025 **Research Assistant (part-time)**

University of Sheffield · Sheffield, UK

- Developed LLM-based multi-agent decision-support methods for Pezego, a mobile pest-management tool for agronomists and farmers, and led the resulting preprint.

Sep 2021 - May 2022 **Graduate Teaching Assistant (part-time)**

University of Sheffield · Sheffield, UK

- Delivered laboratory sessions and student workshops and assessed coursework across Machine Learning, Adaptive Intelligence, Text Processing and Natural Language Processing.

Jul 2010 - May 2019 **Control System Development Engineer; Team Leader**

China Nuclear Power Engineering Co. (CNPEC) · Shenzhen, China

- Led a 13-member multidisciplinary control-system design team, coordinating parallel technical deliverables and engineering documentation for nuclear power projects.

EDUCATION

2020 - 2025 **PhD, Computer Science**

University of Sheffield

Efficient and privacy-aware federated learning under data and system heterogeneity: partial-model training, few-shot updates and active data selection. Supervised by Po Yang and Valentin Radu.

2019 - 2020 **MSc, Artificial Intelligence**

University of Edinburgh

Merit.

2008 - 2010 **MSc, Control Science and Engineering**

Harbin Institute of Technology

2004 - 2008 **BEng, Automation**

Harbin Institute of Technology

PUBLICATIONS

2026 **Habitat classification from ground-level imagery using deep neural networks**
Hongrui Shi, Lisa Norton, Lucy Ridding, Simon Rolph, Tom August, Claire M. Wood, Lan Qie, Petra Bosilj, James M. Brown
Ecological Informatics, 95, 103751 · DOI · code

2025 **Federated learning with workload reduction through partial training of client models and entropy-based data selection**
Hongrui Shi, Valentin Radu, Po Yang
IEEE ICDCS, 373-383 · DOI

2023 **Lightweight workloads in heterogeneous federated learning via few-shot learning**
Hongrui Shi, Valentin Radu, Po Yang
DistributedML at ACM CoNEXT, 21-26 · DOI · code

2023 **Distributed training for speech recognition using local knowledge aggregation and knowledge distillation in heterogeneous systems**
Hongrui Shi, Valentin Radu, Po Yang
EuroMLSys at EuroSys, 64-70 · DOI

2022 **Data selection for efficient model update in federated learning**
Hongrui Shi, Valentin Radu
EuroMLSys at EuroSys, 72-78 · DOI

2021 **Towards federated learning with attention transfer to mitigate system and data heterogeneity of clients**
Hongrui Shi, Valentin Radu
EdgeSys at EuroSys · DOI

FUNDING

2026 **HabiFusion and HabiBench - multimodal habitat assessment** [SUBMITTED]
Principal Investigator. Submitted to EPSRC-supported OMAIB Rounds 1 and 2; the second redesigned around open LUCAS data, reliability measures and proposed deployment validation with Mozaic Earth.

TALKS

2025 **Computer vision for habitat monitoring**
NERC Digital Gathering; University of Lincoln RAISE showcase

2025 **AI-Hab stakeholder workshop (co-organiser)**
with Defra and Mozaic Earth

2022 **Closing the gap between client and global model performance in heterogeneous FL**
On-device AI, Samsung R&D Institute UK

2022 **Challenges and solutions in heterogeneous federated learning**
On-device AI, Samsung R&D Institute UK

2022 **Data selection for efficient model update in federated learning**
EuroMLSys at EuroSys

2021 **Towards federated learning with attention transfer**
EdgeSys at EuroSys

SERVICE

ongoing **Peer reviewer**
IEEE ICIP; IEEE Internet of Things Journal; Ophthalmology Science

2021 - 2026 **Teaching and mentoring**
Supervised a master's student; guest teaching on efficient on-device learning and computer vision for habitat monitoring

2022 **Student volunteer**
ACM/IEEE Conference on Information Processing in Sensor Networks (IPSN)